claude-windows / a9fd4426-f234-4e4a-bfb4-1f2c8255c661

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\a9fd4426-f234-4e4a-bfb4-1f2c8255c661\subagents\workflows\wf_076beba8-59c\agent-af0235f1450007a94.jsonl`
- SHA-256: `52b6c0ca24a3925a3a509c9511c318c154ba3ce342626b5778c5f5081199b8b9`
- Source modified: `2026-05-31T04:25:02+00:00`
- Imported at: `2026-07-05T16:48:21+00:00`
- project: `wf_076beba8-59c`
- session_id: `a9fd4426-f234-4e4a-bfb4-1f2c8255c661`

Transcript

1. user (2026-05-31T04:24:04.271Z)

Adversarial Claim Verifier (voter 1/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Identify the best LOCAL (self-hosted, single NVIDIA CUDA GPU) vision / document-AI / OCR models ? and, as a clearly-flagged approval-gated fallback only, paid cloud API services ? for parsing BANK STATEMENT PDFs into structured transaction tables (date, narration/description, withdrawal/debit, deposit/credit, running balance), for a LOCAL-FIRST personal-finance tool ("muneem").

Target documents: Indian bank statements (HDFC, Axis, ICICI, AU Small Finance) ? dense, multi-column, often SEMI-RULED or borderless tables, sometimes multiple accounts per statement; both born-digital (text layer present) and the harder cases where pdfplumber's extract_tables/coordinate clustering is unreliable (e.g. Axis, whose column layout varies between statements).

HARD CONSTRAINTS (a model that violates these is unusable for us):
1. MUST run 100% LOCALLY for statement contents (highly sensitive). NO cloud APIs by default. Paid/cloud is acceptable ONLY as an explicitly-approved fallback ? still report the best cloud options but label them clearly as "cloud/sensitive ? approval-gated, not default".
2. MUST be usable from Python (transformers / vLLM / Ollama / ONNX Runtime / native lib).
3. License MUST be permissive & commercially safe for BOTH the CODE and the MODEL WEIGHTS: MIT, BSD, Apache-2.0, ISC, HPND. AVOID and explicitly FLAG anything GPL/LGPL/AGPL, CC-BY-NC / non-commercial, research-only, or custom "community"/acceptable-use licenses with restrictions. This is critical ? many document-AI models and their training datasets are non-commercial (e.g. LayoutLMv3 weights are CC-BY-NC; Surya and Marker use GPL-or-commercial dual licensing; several VLM weights ship custom restrictive licenses). Verify the WEIGHTS license, not just the repo license.
4. Hardware: a single consumer/prosumer NVIDIA GPU. Give VRAM tiers (~8-12 GB consumer, ~16-24 GB) and which models/quantizations fit each.

Evaluate and compare these LOCAL approaches ? for each give: code license, MODEL-WEIGHTS license (state explicitly), latest release / maintenance status as of 2025-2026, VRAM need + quantization options, Python integration path, and reliability specifically on DENSE FINANCIAL TABLES / multi-column statements:
- Open vision-language models (VLMs) for document/table understanding: Qwen2-VL and Qwen2.5-VL (note per-size license ? which sizes are Apache-2.0 vs restricted), MiniCPM-V 2.6, InternVL2 / 2.5, Microsoft Phi-3.5-vision, Mistral Pixtral, Meta Llama-3.2-Vision (flag the Llama Community License restrictions), Moondream2, GOT-OCR2.0, olmOCR and other OCR-specialized VLMs.
- OCR-specialist / document-AI pipelines: PaddleOCR + PP-Structure (table recognition), docTR, EasyOCR, Tesseract (already used), Microsoft Table Transformer / TATR (check code AND PubTables-1M dataset/weights license), Donut, Surya (FLAG license), Marker (FLAG license),

MinerU / PDF-Extract-Kit, RapidOCR.
- Table-structure recognition specifically (cell/grid reconstruction for borderless tables): TATR, PP-Structure, UniTable, etc.

Also assess: for SEMI-RULED / borderless bank tables, is a prompt-driven VLM (page image -> JSON transactions) more reliable than a classic OCR + table-structure pipeline? What are the accuracy/hallucination risks of VLMs on financial NUMBERS (transposed digits, dropped/duplicated rows), and how should output be VERIFIED? (muneem already machine-checks parses by balance reconciliation ? opening + sum(deposits) - sum(withdrawals) == closing, plus per-row running-balance walks and cross-checks against the bank's printed summary ? so any model's output can be automatically validated, which should inform the recommendation.)

Finally, briefly cover the best PAID/CLOUD fallbacks (clearly flagged approval-gated for sensitive data): AWS Textract (AnalyzeExpense / Tables), Google Document AI, Azure AI Document Intelligence, Reducto, LlamaParse, Anthropic/OpenAI vision ? note any self-hosting option and data-handling terms.

Deliver a RANKED shortlist of the most promising LOCAL options for muneem (NVIDIA GPU, Python, commercial-license-safe), each with license (code + weights), VRAM tier, maturity, strengths/weaknesses for dense bank-statement tables, and a concrete recommendation: which local model(s) to prototype for muneem's statement parsing and how they'd combine with the existing deterministic parsers + balance-reconciliation verification (e.g. VLM as a fallback/cross-check when deterministic parsing fails to reconcile). Cite sources and call out any license uncertainty explicitly.

Claim under review

"Surya's model WEIGHTS are NOT permissively licensed: they are governed by the 'AI PUBS OPEN RAIL-M LICENSE (MODIFIED)', Version 0.1 (March 2, 2023) ? a restricted (OpenRAIL-M family) license, not MIT/BSD/Apache/ISC/HPND. This makes the weights non-compliant with muneem's hard constraint #3 (permissive weights required)."

Source: https://github.com/datalab-to/surya/blob/master/MODEL_LICENSE (primary)
Supporting quote: "AI PUBS OPEN RAIL-M LICENSE (MODIFIED)

Version 0.1, March 2, 2023 (Modified)
http://licenses.ai/"

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

refuted=true if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.
refuted=false ONLY if: claim is well-supported, current, and source quality matches claim strength.
Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-05-31T04:24:06.870Z)

I'll verify this claim about Surya's model weights license. Let me start by fetching the primary source.

3. assistant (2026-05-31T04:24:10.007Z)

I have the tools loaded. Let me fetch the primary source and search for corroborating/contradicting evidence in parallel.

4. user (2026-05-31T04:24:16.730Z)

Surya Model License Analysis

```
**License Name, Version, and Date:**
"AI PUBS OPEN RAIL-M LICENSE (MODIFIED) Version 0.1, March 2, 2023 (Modified)"

**License Type:**
This is a **restricted license** with use-based restrictions, not a permissive open-source
license like MIT or Apache.

**Commercial-Use Restrictions:**
Yes, it contains explicit commercial-use restrictions. Section III, paragraph 5 and Attachment A
specify that users cannot employ the model if their organization:
- Generated over $5,000,000 in gross revenue in the prior year (except for personal/research
use)
- Raised over $5,000,000 in total equity or debt funding (except for personal/research use)
- Offers competing products or services

**Key Provision:**
The license requires that "use-based restrictions in paragraph 5 MUST be included as an
enforceable provision" in any downstream distribution or usage agreements. This means these
restrictions propagate to all derivative works and subsequent users.

The inclusion of "RAIL" in the nomenclature deliberately signals revenue and use-based
restrictions designed to prevent certain commercial applications.
```

5. user (2026-05-31T04:24:17.984Z)

```
Web search results for query: "Surya OCR datalab model weights license OpenRAIL GPL commercial"
```


```
Links: [{"title":"GitHub - datalab-to/surya: OCR, layout analysis, reading order, table
recognition in 90+ languages · GitHub","url":"https://github.com/datalab-
to/surya"},{"title":"surya/MODEL_LICENSE at master · datalab-
to/surya","url":"https://github.com/datalab-
to/surya/blob/master/MODEL_LICENSE"},{"title":"surya/README.md at master · datalab-
to/surya","url":"https://github.com/datalab-to/surya/blob/master/README.md"},{"title":"Surya:
The Revolutionary OCR Toolkit for 90+
Languages","url":"https://converter.brightcoding.dev/blog/surya-the-revolutionary-ocr-toolkit-
for-90-languages"},{"title":"surya-ocr · PyPI","url":"https://pypi.org/project/surya-
ocr/"},{"title":"GitHub - datalab-to/surya: OCR, layout analysis, reading order, table
recognition in 90+ languages","url":"https://github.com/datalab-
to/surya/tree/master"},{"title":"Datalab - Intelligent Document Processing: IDP-
Software","url":"https://idp-software.com/vendors/datalab/"},{"title":"Datalab develops small AI
models to power 'document intelligence' in the
enterprise","url":"https://www.forkable.io/p/datalab-develops-small-ai-models"},{"title":"surya-
ocr 0.17.1 on PyPI - Libraries.io - security & maintenance data for open source
software","url":"https://libraries.io/pypi/surya-ocr"},{"title":"datalab-to/marker |
zdoc.app","url":"https://www.zdoc.app/en/datalab-to/marker"}]
```


```
Based on the search results, here's what I found about Surya OCR's licensing:
```

Model Weights License

```
The model weights use a modified AI Pubs Open Rail-M license (free for research, personal use,
and startups under $5M funding/revenue). However, there's a discrepancy in different
sources?some documents reference startups under $2M funding/revenue as the threshold.
```

Code License

```
The Surya code is licensed under Apache 2.0. However, other sources indicate the code is GPL.
```

Commercial Licensing

For broader commercial licensing of the model weights, visit our pricing page. Additionally, for broader commercial licensing or to remove GPL requirements, visit our pricing page.

The licensing structure appears to be designed to allow free use for research, personal projects, and smaller startups, while requiring commercial licenses for larger enterprises that want to use the model weights or avoid GPL code obligations.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-05-31T04:24:21.093Z)

The primary source confirms the claim with strong specificity. Let me verify the README to confirm Surya's own characterization of the weights license, and check the actual model weights repository on HuggingFace to ensure the claim isn't outdated.

7. user (2026-05-31T04:24:27.189Z)

Licensing Information from Surya README

The README contains a dedicated "Commercial usage" section that addresses licensing:

Code License:
"The Surya code is licensed under Apache 2.0."

Model Weights License:
"The model weights use a modified AI Pubs Open Rail-M license (free for research, personal use, and startups under $5M funding/revenue)."

Commercial Licensing:
"For broader commercial licensing of the model weights, visit our pricing page."

The README explicitly mentions OpenRAIL (as "Open Rail-M"), Apache licensing, and includes a commercial threshold of $5M in funding or revenue for startups seeking free use of the model weights.

8. user (2026-05-31T04:24:30.776Z)

Web search results for query: "Surya OCR license 2025 GPL OpenRAIL weights commercial use restriction datalab"

Links: [{"title":"GitHub - datalab-to/surya: OCR, layout analysis, reading order, table recognition in 90+ languages · GitHub","url":"https://github.com/datalab-to/surya"},{"title":"surya/MODEL_LICENSE at master · datalab-to/surya","url":"https://github.com/datalab-to/surya/blob/master/MODEL_LICENSE"},{"title":"surya-ocr · PyPI","url":"https://pypi.org/project/surya-ocr/"},{"title":"GitHub - CibinQuadance/surya-OCR: OCR and line detection in 90+ languages · GitHub","url":"https://github.com/CibinQuadance/surya-OCR"},{"title":"surya-ocr 0.17.1 on PyPI - Libraries.io - security & maintenance data for open source software","url":"https://libraries.io/pypi/surya-ocr"},{"title":"Surya: The Revolutionary OCR Toolkit for 90+ Languages","url":"https://converter.brightcoding.dev/blog/surya-the-revolutionary-ocr-toolkit-for-90-languages"},{"title":"pitapo/surya · Hugging Face","url":"https://huggingface.co/pitapo/surya"},{"title":"datalab-to/surya-ocr-2 · Hugging Face","url":"https://huggingface.co/datalab-to/surya-ocr-2"},{"title":"Datalab develops small AI models to power 'document intelligence' in the enterprise","url":"https://www.forkable.io/p/datalab-develops-small-ai-models"},{"title":"Surya OCR | tasarim.ai","url":"https://tasarim.ai/en/models/surya-ocr"}]

Based on the search results, here's what I found about Surya OCR's licensing in 2025:

License Structure

The Surya code is licensed under Apache 2.0, while the model weights use a modified AI Pubs Open Rail-M license (free for research, personal use, and startups under $5M funding/revenue).

Commercial Use Restrictions

The underlying model weights are licensed for non-commercial use only, meaning that while anyone can inspect, modify, and redistribute the code subject to GPL license terms, they can't use the trained models in any commercial setting without a paid license.

GPL Requirements

The code itself is GPL-licensed, fostering community contributions while offering commercial licensing options for enterprises that need to remove GPL requirements. To remove the GPL license requirements for inference or use the weights commercially over the revenue limit, contact surya@vikas.sh for dual licensing.

Commercial Licensing Options

For broader commercial licensing of the model weights, Datalab has a pricing page available. Additionally, larger companies need a commercial license from Datalab to remove GPL code requirements and access enterprise support.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

9. assistant (2026-05-31T04:24:45.692Z)

The evidence is consistent and strong across multiple independent sources. Let me verify the claim's specific framing ? the claim says weights are "AI PUBS OPEN RAIL-M LICENSE (MODIFIED), Version 0.1 (March 2, 2023)" and non-permissive. All sources confirm this. The claim is well-supported and current (the MODEL_LICENSE file is on master branch as of now, and the README on master confirms the OpenRAIL-M characterization).

Let me note one nuance for my evidence: the search results introduce some confusion about whether the *code* is Apache-2.0 or GPL. But the claim under review is strictly about the **weights** ? and on that point every source agrees: the weights are governed by the modified AI Pubs OpenRAIL-M license, which is non-permissive (use-based + revenue-threshold restrictions). The claim does not over-extend to the code license.

Let me run through the checklist:

1. **Supported by quote?** Yes. The supporting quote ("AI PUBS OPEN RAIL-M LICENSE (MODIFIED), Version 0.1, March 2, 2023") is verbatim from the MODEL_LICENSE file, and WebFetch on the raw file confirms it contains use-based restrictions (Section III/Attachment A: $5M revenue/funding caps, no competing products). The claim that this is "not MIT/BSD/Apache/ISC/HPND" and "restricted (OpenRAIL-M family)" is directly accurate.

2. **Contradicting evidence?** None found. Surya's own README on the master branch explicitly states "The model weights use a modified AI Pubs Open Rail-M license (free for research, personal use, and startups under $5M funding/revenue)." This is the maintainer's own characterization ? it corroborates, doesn't contradict.

3. **Source quality?** Primary (the license file itself + the project's own README). Highest possible quality for a licensing claim.

4. **Outdated?** No. Both the MODEL_LICENSE and README are fetched from the current `master` branch. The OpenRAIL-M weights license is the current, live licensing as of 2025-2026.

5. **Marketing/speculation?** No ? it's a license document and official README, not a benchmark or forum post.

The only minor imprecision: the version-0.1 license text dates to March 2, 2023, but

Surya/datalab applies it to current weights ? this is the license *version*, not a claim that the license is stale. The claim correctly attributes it as the governing license. This does not undermine the claim.

The claim under review is well-supported, current, and backed by primary sources (the license file and the project's own README). It does NOT overreach into the disputed code-license territory. It accurately concludes the weights are non-compliant with a "permissive weights required" constraint.

I cannot refute this claim.

10. user (2026-05-31T04:24:57.288Z)

Structured output provided successfully

11. assistant (2026-05-31T04:25:02.071Z)

I've completed my adversarial verification of the claim.

Verdict: refuted = false (claim confirmed, could not refute).

The claim that Surya's model weights are governed by the "AI PUBS OPEN RAIL-M LICENSE (MODIFIED), Version 0.1 (March 2, 2023)" ? a restricted, non-permissive license ? is well-supported by primary sources. I verified the actual MODEL_LICENSE file on the current master branch (matches the supporting quote verbatim, contains $5M revenue/funding caps and use-based restrictions), and Surya's own README independently corroborates the OpenRAIL-M weights characterization. Source quality is the highest possible (primary license file + official README), the claim is current (live master branch), and it correctly scopes itself to the weights without overreaching into the separate code-license ambiguity. The conclusion that these weights violate muneem's hard constraint #3 (permissive weights required) is sound.